

EHSAN NOURI

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Summary

Mechanical R&D Engineer specializing in precision mechanical design, experimental prototyping, and scientific instrumentation. Experienced in CAD-based mechanical development, Additive Manufacturing, FEA/CFD, and hands-on fabrication, with a strong focus on rapid prototyping, experimental testing, and iterative design optimization. Skilled in developing and validating mechanical and electromechanical systems, integrating sensors and hardware, and translating engineering concepts into functional laboratory prototypes.

Work Experience

Deggendorf Institute of Technology, Germany, *Student Assistant* *July 2026 – Present*

- Co-authored technical manuscripts by developing experimental test setups, conducting laboratory investigations, and acquiring reliable empirical data.
- Characterized mechanical behavior and structural performance through iterative experimental testing and Finite Element Method (FEM) simulations.

Pouyesh Mechatronics, Iran, *Mechanical Design Engineer* *Oct 2024 – Feb 2026*

- Designed precision mechanical housings for optical gyroscope systems using CATIA, integrating optical components with mechanically stable and manufacturable structures.
- Generated manufacturing-ready technical drawings applying Geometric Dimensioning and Tolerancing (GD&T) principles to ensure dimensional accuracy and functional interfaces.
- Managed external fabrication workflows including CNC machining and Additive Manufacturing, and validated manufactured components through dimensional metrology and inspection.

Institut für Prototypen und Modelltechnik – IPM, Germany, *Research Assistant* *Oct 2023 – Sep 2024*

- Investigated critical FDM process parameters in novel needle-insertion 3D printing to improve inter-layer adhesion, dimensional consistency, and structural performance.
- Conducted tensile and strain analyses of engineering polymers, combining experimental characterization with quantitative data analysis to evaluate mechanical performance.

Institut für Prototypen und Modelltechnik – IPM, Germany, *Master Thesis* *Apr 2023 – Sep 2023*

- Optimized extrusion mechanics and hot-end thermal stability of the Mycusini 1.0 system through mechanical development, experimental testing, and iterative prototype refinement.
- Refined system firmware and integrated electromechanical test feedback to improve motion control, thermal performance, and overall system reliability.

Institut für Prototypen und Modelltechnik – IPM, Germany, *Internship* *Oct 2022 – Mar 2023*

- Prototyped an experimental FDM 3D printer system, performing mechanical assembly, hardware integration, and custom firmware configuration.
- Conducted statistical analysis of material strain data to experimentally validate the mechanical performance and repeatability of a newly developed needle-insertion nozzle.

Atin Co, Iran, *Mechatronics Engineer* *Jan 2018 – Sep 2021*

- Designed, manufactured, and implemented a satellite transportation container for the safe transportation of satellites, with full responsibility for mechanical design, structural architecture, fabrication, and integration of mechanical and electronic subsystems.
- Designed and fabricated passive Wire Rope Isolators (WRI) according to specified performance requirements and MIL-STD-810 vibration and shock requirements; experimentally characterized their shock-isolation performance using a magnetic shock testing system.
- Designed and built a 2-DOF helicopter test unit for experimental investigation of yaw and pitch dynamics, including mechanical design, actuation, instrumentation, and electromechanical system integration.

- Designed and fabricated an aerodynamic propeller test unit for experimental evaluation of thrust force and aerodynamic performance.
- Performed 3D CAD design, precision mechanical fabrication, electromechanical integration, rapid prototyping, and experimental validation of specialized engineering systems in accordance with applicable MIL and industry standards.

- Designed and integrated a prototype hybrid solar–wind power generation system, covering mechanical structure, component integration, and electromechanical subsystem development.

***Spad3D, Iran**, Mechanical Engineer* *Jun 2013 – Oct 2017*

- Assembled and calibrated custom high-precision FDM 3D printers, optimizing mechanical hardware, structural alignment, and multi-axis motion performance.
- Programmed and debugged open-source Marlin firmware to improve multi-axis motion control, thermal management, and system reliability.

Education

Deggendorf Institute of Technology, Germany *March 2026–Present*

- MSc. Eng. Mechatronics and Cyber Physical System

Coburg University of Applied Science, Germany *Oct 2021–Sep 2023*

- MSc. Eng. Analytical Instruments, Measurement and Sensor Technology
- **Thesis:** Optimization of Chocolate 3D Printer
- **Supervisor:** Prof. Dr. Markus Stark

Azad University of Karaj,Iran *Oct 2008–Feb 2013*

- BSc. Eng. Mechanical Engineering
- Teaching Assistant for some courses: Static, Mechanics of Materials, Industrial Drawing

Technical Skills

- **CAD / CAM Systems:** CATIA, SolidWorks, AutoCAD, Cura
- **Simulation & Analysis (CAE):** ANSYS (FEA/CFD), Structural Analysis, Fluid Dynamics
- **Prototyping & Manufacturing:** FDM 3D Printing, CNC Laser Machining, Rapid Prototyping
- **Instrumentation:** Advanced Measurement Systems, Sensor Interfacing, Electromechanical Prototyping
- **Testing & Reliability:** FMEA, Experimental Testing, Failure Analysis, System Troubleshooting

Languages

- **German:** A2
- **English:** Fluent
- **Persian:** Native
- **Azerbaijani:** Fluent

Certificates

- **SolidWorks:** CSWA, CSWA-AM, CSWA-xDesign, CSWA-xShape, CSWA-xMold, CSWA-E, CSWA-SD [Dassault Systèmes], 2026
- **CATIA V5:** Advanced [ParsPajouhan Inst.], 2020
- **Ansys:** Intermediate [Sharif University], 2019
- **Abaqus:** Intermediate [Sharif University], 2019
- **Python:** Intermediate [Sematec], 2024
- **Driving Licence:** AM/B/L, 2024

Ehsan Nouri

Deggendorf, November 1, 2026